



A novel LiDAR sensor alignment inspection system for automobile productions using 1-D photodetector arrays

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ABSTRACT

This paper proposes a novel system to inspect the LiDAR sensor misalignment on a vehicle during the sensor assembly process in the autonomous vehicle productions. The inspection system is primarily designed for deployment in the factory-level assembly process for commercial vehicles with ADAS and autonomous driving functions. To obtain sub-degree and millimeter accuracy levels of the LiDAR sensor alignment or the extrinsic pose from the vehicle body, this study proposed a new concept of LiDAR sensor extrinsic calibration method using a target board with embedded photodetector(PD) arrays, named the PD-target system. Furthermore, the proposed system requires only the simple design of the target board at the fixed pose in a close range to be readily applicable in the automobile manufacturing environment. From the estimation of the extrinsic pose of the LiDAR sensor from the target board, the LiDAR sensor alignment with respect to the vehicle body can be easily derived in the controlled factory environment, where positions of both the target board and the vehicle are assumed to be known. The experimental evaluation of the proposed system on low-resolution LiDAR sensor showed that the LiDAR sensor offset pose can be estimated within 0.1 degree and 3 mm levels of precision. The high accuracy and simplicity of the proposed calibration system make it practical for production of reliable and safe autonomous transportation.

1. Introduction

LiDAR(light detection and ranging) system is an essential perception sensor, along with camera and radar, to provide 3D data of their surroundings for smart mobility systems such as vehicles, unmanned aerial systems and robots. LiDAR sensor perceives the surroundings with a point cloud of 3D data that is used to generate a high-resolution 3D map and detect and track multiple objects such as vehicles and pedestrians. If the application of LiDAR sensor expands on automobiles for mass production, an important safety issue regarding the sensor misalignment can arise.

A LiDAR sensor is mounted on intelligent transportation, for locating multiple targets and obstacles [1–3] for tasks such as collision avoidance and path planning [4], and generating 3D maps of the surroundings for localization [5–8]. To generate accurate reconstruction and interpretation of the surroundings, precise assembly of LiDAR sensor on mobile platforms is critical. An alignment offset as low as a few degrees could cause a significant error in predicting the position of obstacles relative to the ego-vehicle and can result in dangerous situations.

Therefore, a systematic LiDAR sensor misalignment inspection for a commercial autonomous vehicle is necessary because an unexpected offset in the sensor assembly could reduce the reliability and safety of driving [9].

Towards the practical application in the automotive industry, the LiDAR sensor inspection system needs to be integrated in the existing sensor assembly and inspection process of manufacturing commercial vehicle. Considering the actual in-factory environment of the automobile production process, the inspection area can be assumed to be confined to a limited space of a few meters that includes complex machinery for various assembly tasks, such as rails and robots. Also, the sensor inspection system should be able to give accurate misalignment measurement even on low-resolution LiDAR sensors, which could be deployed as auxiliary sensors on autonomous vehicles.

For this reason, this paper assumes the in-factory inspection for the LiDAR sensor calibration is constrained that restricts the setup of the calibration target within a few meters. Also only artificial target is allowed for the flat ground or wall may not be available as due to other existing infrastructures. In addition, the vehicle and the calibration

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target are assumed to be static motion during the inspection time, in contrast to other on-drive calibration methods.

Therefore, this paper aims to design an automatic LiDAR sensor alignment inspection system for commercial vehicles that can be applied to the in-factory sensor assembly process of the aforementioned restrictions and environment. For safety consideration, the alignment system should be designed to satisfy high level of precision.

The sensor alignment inspection is based on the extrinsic parameter calibration that estimates the pose of LiDAR sensor with respect to the vehicle body or the calibration target that is at the known position from the vehicle, as in the controlled factory-level environment. There are several approaches for the LiDAR sensor extrinsic calibration. One main approach is solving for the geometric relationship with respect to other perception sensor such as a camera [10–15]. Other approach is boresight calibration [16–19] that estimate the orientation of the LiDAR sensor by scanning landmarks [20–25]. This approach of LiDAR sensor extrinsic calibration can use the flat ground plane and a long pole from a series of poses [20], the ground as the target object at various poses [21] or multiple planar objects such as walls and floor [22,26] to estimate the rigid transformation between the LiDAR sensor and the target objects.

These previous extrinsic calibration methods require calibration targets scanned at different poses by using multiple targets or moving the LiDAR sensor. Thus, these methods are limited to be applied to the actual in-factory environment where both the target and the vehicle are both static during the inspection and there is usage restriction of multiple calibration targets.

A planar target based method was proposed [9] to satisfy the aforementioned in-factory vehicle production. However, the main drawback of that method is the position uncertainty of the feature beam points due to low resolution of the LiDAR sensor, as explained in Section 3.

Therefore, to overcome these aforementioned limitations, this paper proposes an extrinsic calibration method that uses only single calibrating target, at a close range and requires only one pose of the target scan. Most importantly, the misalignment pose is estimated with an accuracy within a few millimeters and at the sub-degree level even for a low-resolution mobile LiDAR sensor. Since the position of the calibration target and the vehicle frame are assumed to be known in the factory-level environment, the LiDAR sensor alignment inspection becomes the problem of estimating the pose of the LiDAR sensor from the target board, without loss of generality.

The objective of the paper is to solve the uncertainty issue of the method in [9] by utilizing photodetector(PD) arrays to search and position the target features more accurately. For this, a simple planar target with embedded photodetector arrays, named as the PD–target system. Modules of NIR(Near Infra-red) photodetector 1D arrays, are arranged near the corners of a planar mid-sized target, to detect the position of the beam spot on the target surface. The positions of the corresponding beam points measured from the LiDAR sensor point cloud data and the target photodetectors are used for the initial pose estimation. Then, an iterative nonlinear optimization method is applied to obtain the final pose of the LiDAR sensor with respect to the target body frame, which, without loss of generality, is assumed to be the vehicle body frame.

The contributions of this paper can be summarized as follows.

First, the proposed system is a novel in-factory calibration method that combines a target board with photodetector arrays (PD–target) that has not been proposed in other literature. Second, the system can estimate the LiDAR sensor alignment with the improved accuracy and precision compared to our previous work [9].

Last, this system requires only a single target board scanned at the static pose that makes it applicable to the actual manufacturing and sensor assembly process of the commercial automobiles.

The paper is organized as follows. In Section 2, related work is reviewed and compared with this study. In Section 3, the LiDAR sensor measurement uncertainty is analyzed that affect the accuracy of pose

estimation. In Section 4, the design process of the proposed PD–target system is described, and the signal processing of the photodetector and the LiDAR sensor are explained. The pose estimation algorithm is explained in Section 5, which is followed by experiments in Section 6 to validate the overall accuracy and precision of the alignment estimation.

2. Related work

LiDAR sensor calibration can be classified as intrinsic parameter calibration and extrinsic parameter calibration. The intrinsic calibration refers to optimizing the internal parameters to minimize the systematic error of the LiDAR sensor measurements due to the optical parts and conversion from raw data to 3D points [22,29,30]. However, this paper focused on the extrinsic calibration of LiDAR sensor.

Research on LiDAR sensor extrinsic calibration are mainly focused on estimating the geometric relationship of the LiDAR sensor with respect to the camera [10–12]. The extrinsic parameters are estimated by extracting the 3D–2D correspondences from the point cloud of 3D data and 2D image pixels using one or multiple planar targets [13,31], of various such as circular [14,32], trihedral [33] and 3D cubic object [15]. A common method is using a chessboard at multiple poses to determine the correspondence of the planes and lines and applying an optimization method [34–36].

The main approach of the LiDAR sensor extrinsic calibration with respect to a reference frame is based on a plane-based method. The LiDAR sensor scans surfaces of the natural or artificial planar objects to estimate the geometric constraints of the plane model that are needed in minimizing the calibration errors [16,20–25]. The extrinsic calibration proposed by [20] first computes the pitch and roll angles from the flat ground plane and then computes the navigation angle by using a long pole from a series of poses with on-vehicle IMUs and GPS. A similar method of calibrating multiple 2D laser scanners using flat ground and vertical pole was also proposed in [37]. Another extrinsic parameter methods of a 3D LiDAR sensor used a calibration object consisting of four walls and the floor [22]. After estimating the sensor intrinsic parameters, the extrinsic rigid transformation between the LiDAR sensor and the target objects was estimated with an optimization method. Similar approach that used multiple planar objects of walls, floor and other artificial planar objects are proposed in [23–25]. These methods are run dynamically with the vehicle moving to scan the objects at multiple poses. However, the vehicle and the target objects could be restrained in static motion for the in-factory calibration environment.

A static LiDAR sensor inspection based on a single target board was proposed [9] to satisfy the in-factory environment of vehicle production. It used only a single board at a close range and showed measurement accuracy of 0.2degree and 4 mm. However, the main drawback is the measurement uncertainty caused by the target corners detection with low-resolution LiDAR sensor scans, as explained in Section 3. This paper solved the uncertainty issue by applying photodetector arrays to search and position the target features more accurately.

Instead of using planar models, the feature-based methods use sparse feature points such as corners and edges that are extracted from the scanning the objects to estimate the LiDAR sensor pose. A method of estimating the relative pose between the 2D LiDAR sensors and the object [27] extracted four edge points of the object as the feature points. Then, the positions of these feature points acquired from the LiDAR sensor measurements were used to solve for the relative pose. Other similar methods [38–40] used a single board and three of laser beam points with the aid of a camera to estimate the relative 6-DOF pose.

The comparison of selected previous methods are summarized in Table 1. The dynamic method refers the LiDAR sensor is continuously scanning the objects as the vehicle is moving around. Almost all of the previous methods require multiple targets or a target from multiple

Table 1
Comparison of LiDAR sensor extrinsic calibration methods.

Method	LiDAR	Target type	Ext. Sensor	Motion	Precision
Kim et al. [27]	2Dx2	Single board	–	Static	≤ 13 mm, 2°
Atanacio-Jimenez et al. [22]	3D	4* walls & floor	–	Static	≤ 14.9 mm
LeScouarnec [26]	3D	Vertical & horizontal plane	INS	Static	$\leq 0.032^\circ$
Zhu et al. [20]	3D	Ground & series of poles	IMU, GPS	Dynamic	≤ 50 mm
Hillemann et al. [28]	3D	Multiple landmarks	IMU, GPS	Dynamic	millidegree order
Heinz et al. [23,24]	2D	Multiple landmarks	GNSS, IMU	Dynamic	≤ 4.5 mm, 0.02°
Diehm et al. [16]	3Dx2	Ground	GNSS, IMU	Dynamic	$\leq 0.5^\circ$
Oh et al. [9]	3D	–	Single board	Static	≤ 4.9 mm, 0.19°
Ours	3D	Single board	Photodetectors	Static	≤ 2.6 mm, 0.1°

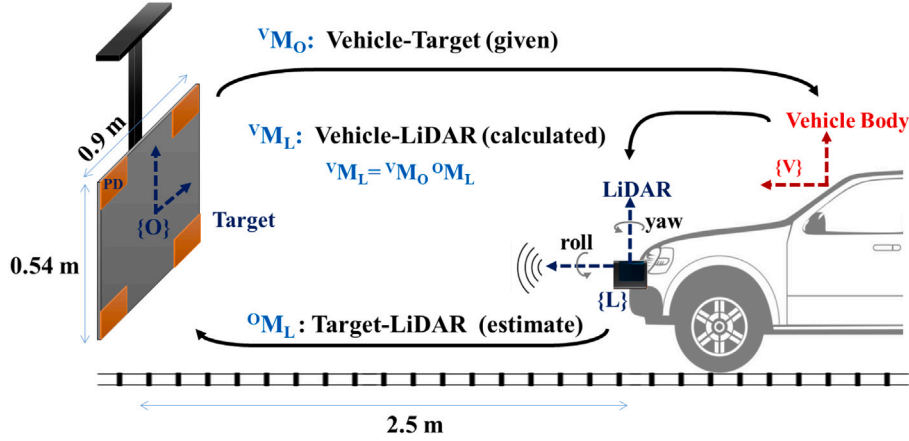


Fig. 1. Overview of LiDAR extrinsic calibration using PD-target board. The Vehicle-LiDAR pose can be calculated from estimating the Target-LiDAR pose and using the known Vehicle-Target pose.

views. Considering the application in the commercial mobility manufacturing, the extrinsic calibration method should use a single and compact target fixed at one static pose, should not use the ground or any other on-site structures, should not be high in computational burden and must estimate the LiDAR sensor misalignment with an accuracy of a few millimeters and at the subdegree level.

Since these methods have limitations to satisfy the aforementioned in-factory calibration application, this paper proposes a new extrinsic calibration method using a static method using a single target with photodetector arrays named as the PD-target system.

3. Analysis of LiDAR measurement error

The overview of the proposed PD-target system for the LiDAR sensor alignment inspection is shown in Fig. 1. The overall system is similar to the LiDAR inspection system [9] that used only a single target board.

This section discusses the major drawback of that method [9] in terms of inaccurate measurement of beam positions. Then, we propose our method to measure those beam positions with higher precision by applying photodetectors (PD) on the target surface.

3.1. Resolution of 3D measurement

The LiDAR sensor model used in this paper is Velodyne VLP-16, which has 16 vertical channels of LEDs emitting 903 nm near-infrared (NIR) wavelength. The firing of beam pulses is a series of a short bursts with a period of 2.3 μ s. Therefore, the minimum speed of ADC (analog-digital converter) for capturing a LiDAR sensor pulse should be at least 1 MHz.

From the raw LiDAR sensor data, the beams on the target board can be defined to be bounded by the vertical angle ranging from ω_1 to ω_n and the azimuth angle from α_1 to α_m , where the values of n and m are the maximum numbers of scan channels and the number of beams per

channel on the given target board dimension, respectively, as described in Fig. 2.

VLP-16 has an azimuth and vertical angular resolution of 0.2 and 2 degrees, respectively, and has the H_{res} of 9 mm and V_{res} of 87 mm, on the target board located at 2.5 m of distance.

3.2. Uncertainty of 3D measurement

The principle in estimating the relative pose between the LiDAR sensor and the target is to find the correspondence of feature points, ${}^O P$ and ${}^L P$, which are measured from the two different coordinate frames of the LiDAR sensor body, $\{L\}$, and the target system, $\{O\}$, respectively, as shown in Fig. 2. The relationship between the two groups of correspondence feature points, ${}^O P$ and ${}^L P$, describes the orientation matrix ${}^O R_L$ and the translation vector ${}^O T_L$ of the LiDAR sensor with respect to the reference target board, ${}^O M_L$.

To solve for the transformation matrix ${}^O M_L$, the 3D positions of feature beam points measured by the target body frame (${}^O p_i$) and measured in the LiDAR sensor frame (${}^L p_i$) are necessary.

$${}^O P = {}^O M_L P = [{}^O R_L \quad {}^O T_L] {}^L P \quad (1)$$

where the rotation and translations of the LiDAR sensor with respect to the target frame are

$${}^O R_L = R_z(\phi) R_y(\theta) R_x(\psi), \quad {}^O T_L = [\Delta X, \Delta Y, \Delta Z]^T$$

Typically, corners and edges are used as the features of a target for LiDAR sensor calibration. Let the corners of the target board, ${}^O P_{cor,i}$, be the features that need to be detected by LiDAR sensor, as indicated in Fig. 2. For the ideal measurement of that feature point, ${}^L \bar{p}_{i,k}$, the LiDAR sensor should project a beam spot directly on that corner.

$${}^L \bar{p}_{i,k} = (r_{i,k}) \begin{bmatrix} \cos(\omega_{i,k}) \sin(\alpha_{i,k}) \\ \cos(\omega_{i,k}) \cos(\alpha_{i,k}) \\ \sin(\alpha_{i,k}) \end{bmatrix} \quad (2)$$

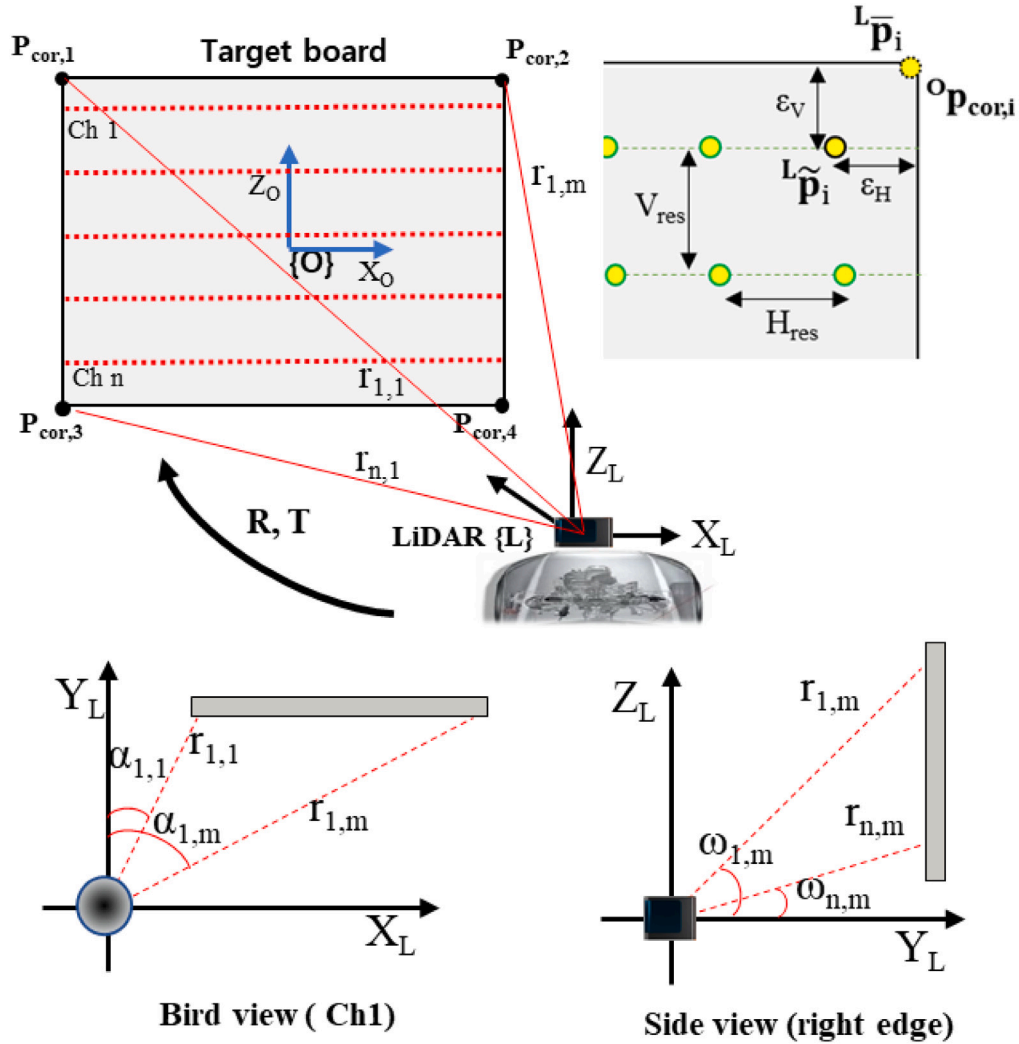


Fig. 2. The geometric relationship of 3D rotation (R) and translation (T) between the target body frame and the LiDAR sensor frame are shown. The number of scan channels of LiDAR sensor is n , and each beam with index (i,j) is measured in terms of the range (r), azimuth(α) and vertical angle (ω).

However, the actual position of the beam closest to the corner, $L\tilde{\mathbf{p}}$, is restricted by the sensor resolution

$$L\tilde{\mathbf{p}}_{i,k} = (\bar{r}_{i,k} + \epsilon_{i,k}) \begin{bmatrix} \cos(\bar{\omega}_{i,k} + \delta\omega_k) \sin(\bar{\alpha}_{i,k} + \delta\alpha_k) \\ \cos(\bar{\omega}_{i,k} + \delta\omega_k) \cos(\bar{\alpha}_{i,k} + \delta\alpha_k) \\ \sin(\bar{\alpha}_{i,k} + \delta\alpha_k) \end{bmatrix} \quad (3)$$

where $\epsilon_{i,k}$ is the depth measurement noise, and $\delta\omega$ and $\delta\alpha$ are the uncertainty of the azimuth and the vertical angle resolution, respectively. The subscript i is the point closest to the i th corner, and k is the scan number.

The deviation of the beam point from the corner of the target surface is bounded by the depth measurement noise ($\epsilon_{i,k}$) in terms of the azimuth and vertical angle resolution as

$$L\bar{\mathbf{p}}_{i,k} - L\tilde{\mathbf{p}}_{i,k} \leq \epsilon_{i,k} = \begin{bmatrix} r_{i,k} \tan \delta\alpha_k \\ 0 \\ r_{i,k} \tan \delta\omega_k \end{bmatrix} \quad (4)$$

This correspondence uncertainty needs to be minimized to obtain higher accuracy in the relative pose estimation. Since the resolution of the given LiDAR sensor cannot be modified unless a higher performance LiDAR sensor replaces it, the feature on the target board needs to be shifted from the corners (${}^OP_{cor}$) to the position where beams are actually projected, (OP_i).

However, this introduces a technical problem of measuring the position of OP_i by the target board. This uncertainty was the major

limitation for the LiDAR sensor inspection system that used only a single target board in [9].

Thus, this paper proposes a method to measure the position OP_i with high precision by using photodetectors attached on the target surface.

4. Design of PD-target system

4.1. Photodetector array

Based on the analysis of the laser beams in Section 3., the appropriate photodetector should be sensitive to the NIR spectrum and have a bandwidth higher than 1 MHz. In addition, the active sensor area of the photodetector should be large enough to capture the beam spot diameter. This paper selected a photodetector 1-D array of 16 diode elements (Hamamatsu S4111), which is sensitive from the UV to the NIR spectrum. The active sensor area is 16 mm by 1.45 mm and the overall dimension, is shown in Fig. 3.

4.2. PD-target board configuration

The photodetector(PD) 1-D arrays are positioned on the target board to capture the LiDAR sensor beam points, which are near the target edges and on the first and the last vertical laser scans, as shown in Fig. 4.

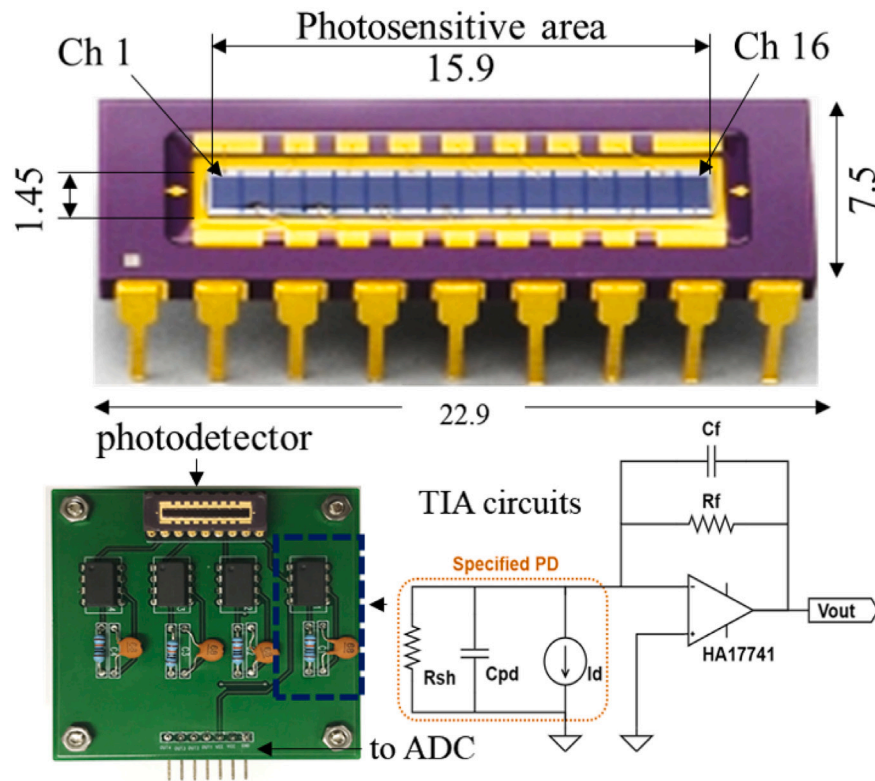


Fig. 3. Photodetector 1-D array of 16 diode elements (Hamamatsu S4111) and the PCB of the signal processing circuit designed in this study.

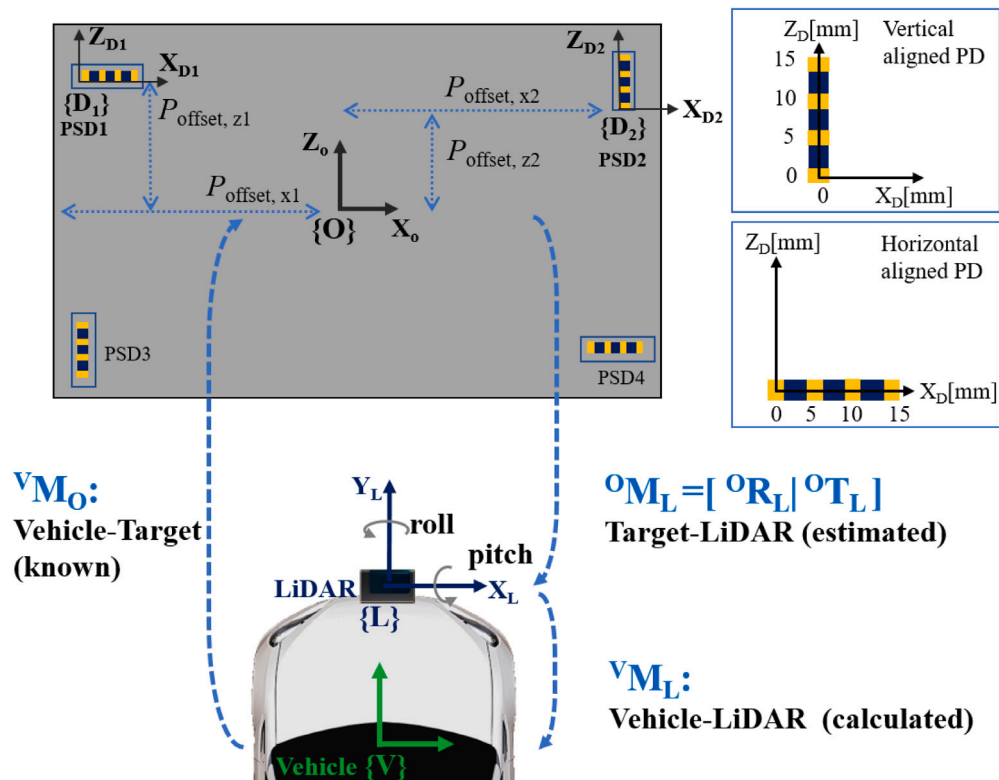


Fig. 4. Overview of LiDAR sensor extrinsic calibration using PD-target board. The Vehicle-LiDAR sensor pose can be calculated from estimating the Target-LiDAR sensor pose and using the known Vehicle-Target pose.

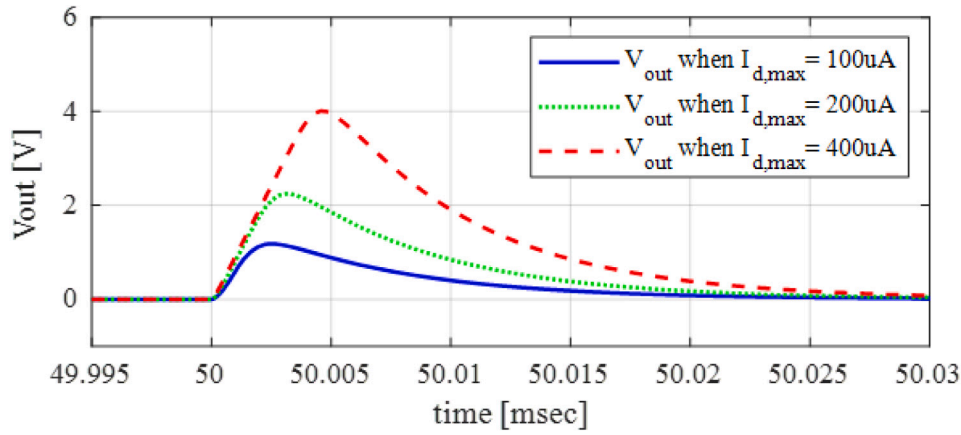


Fig. 5. Simulation of the designed TIA circuit with an input signal of 10-Hz laser pulses with different power intensities that generate current of 100 to 400 μA from the photodetector.

Table 2
OP-AMP and photodetector parameters.

Symbol	Description	Value
GBWP	Op-amp gain bandwidth product	1 [MHz]
A_{OL}	Op-amp DC open-loop gain	106 [dB]
$C_{i,amp}$	Op-amp input capacitance	1.4 [pF]
C_{pd}	Photodiode junction capacitance	200 [pF]
R_{sh}	Photodiode shunt resistance	250 [G Ω]

Since each photodiode is a 1D array, it can detect the center position of a single beam spot in its own coordinate frame, $\{D\}$. The beams photodiodes are referred to as the key beam points throughout the paper.

Then, the position of the beam in the target coordinate frame $\{O\}$ can be easily converted from $\{D\}$ with the matrix $P_{offset} = [X_{pd,offset}, Z_{pd,offset}]$ that describes the location of each photodetector attachment from the center of the board.

$${}^O\mathbf{P} = {}^D\mathbf{P} - \mathbf{P}_{offset} \quad (5)$$

The 1D PD could be attached to the target surface at any angle. However, we considered the horizontal and vertical orientations and compared the performance of these two configurations.

4.3. Signal processing circuit

The photodetector diode generates an electric current when the LiDAR sensor beam is projected on it. The output of the PD should be pre-amplified voltage to be used as the source for the ADC that is connected to the main processor. Thus, the trans-impedance amplifier (TIA) circuit is designed to convert the current to voltage and amplify the output to the usable level. The designed TIA circuit consists of an OP-amplifier, a feedback resistor, and a capacitor, as shown in Fig. 3 (see Table 2).

The value of the feedback resistor, R_f , is determined from the ratio of the maximum photodiode current and the preferred range of the output voltage. The electric current of the PD generated by the laser photons from 2.5 m is expected to be a maximum of 100 μA .

Targeting the output voltage to be within the range of 0 V to 10 V, the R_f value was designed to be 100 k Ω .

The voltage out of the TIA circuit can be expressed as a first-order system as

$$V_{out} = \frac{I_d R_f}{(R_f C_f s + 1)} \quad (6)$$

However, to avoid any unwanted oscillation in the signal, the stability of the circuit needs to be analyzed with the noise gain, which is the

transfer function of the 1st order system of one zero and pole, composed of all resistors and capacitors in the amplifier and the photodetector.

$$G_{noise}(s) = \frac{(R_f + R_{sh})}{R_{sh}} \frac{\left(\frac{R_f R_{sh}}{R_f + R_{sh}}\right)(C_f + C_{in})s + 1}{R_f C_f s + 1} \quad (7)$$

where C_{in} is the combination of capacitors at the input of the amplifier. All the parameters except the variable R_f and C_f are predetermined by the parts specification.

The oscillation or the stability of the signal can be predicted with the circuit quality factor (Q), which can be calculated from the TIA circuit phase margin [41].

$$Q = \left(\left(\frac{1}{(\tan \varphi)^2} + 0.5 \right)^2 - 0.25 \right)^{\frac{1}{4}} \quad (8)$$

A Q-factor value below 0.5 causes the system to be overdamped to avoid any oscillation in the output voltage. In this application, the feedback capacitance of 68 pF was selected and resulted in a Q-value of 0.47 that yielded an overdamped system.

Circuit simulation with the designed TIA circuit is conducted by applying 100 ms (10 Hz) of laser pulses with different power intensities to generate the current input ranging from 100 to 400 μA to the circuit.

The results in Fig. 5 showed that the output voltage can be generated as high as 4 V when the current of 400 μA is applied by laser beam. Moreover, there is no oscillation in the output signals, as required. The circuitry of TIA was designed on a PCB that can amplify up to 4 diode channels of a PD array, as shown in Fig. 3.

The sampling rate for each ADC channel should be higher than 2 MHz to capture laser pulses of the 2.2- μs period. In addition, all the ADC channels must be concurrent to capture all the output signals of the PD simultaneously.

Limited by the availability of ADC channels and DAQ boards, only two PDs and four out of sixteen diodes from each PD were sampled simultaneously.

5. Algorithm of LiDAR sensor extrinsic calibration

5.1. Preprocessing

From the whole point cloud of one LiDAR sensor scan, the target board is segmented as the region of interest (ROI). A Euclidean clustering algorithm is applied to segment the planar target automatically from the background. With the point cloud data of the ROI, the best-fit plane model of the target is derived using a least squares method. Then, the ROI beam points are projected onto the fit plane surface to minimize the sensor measurement error.

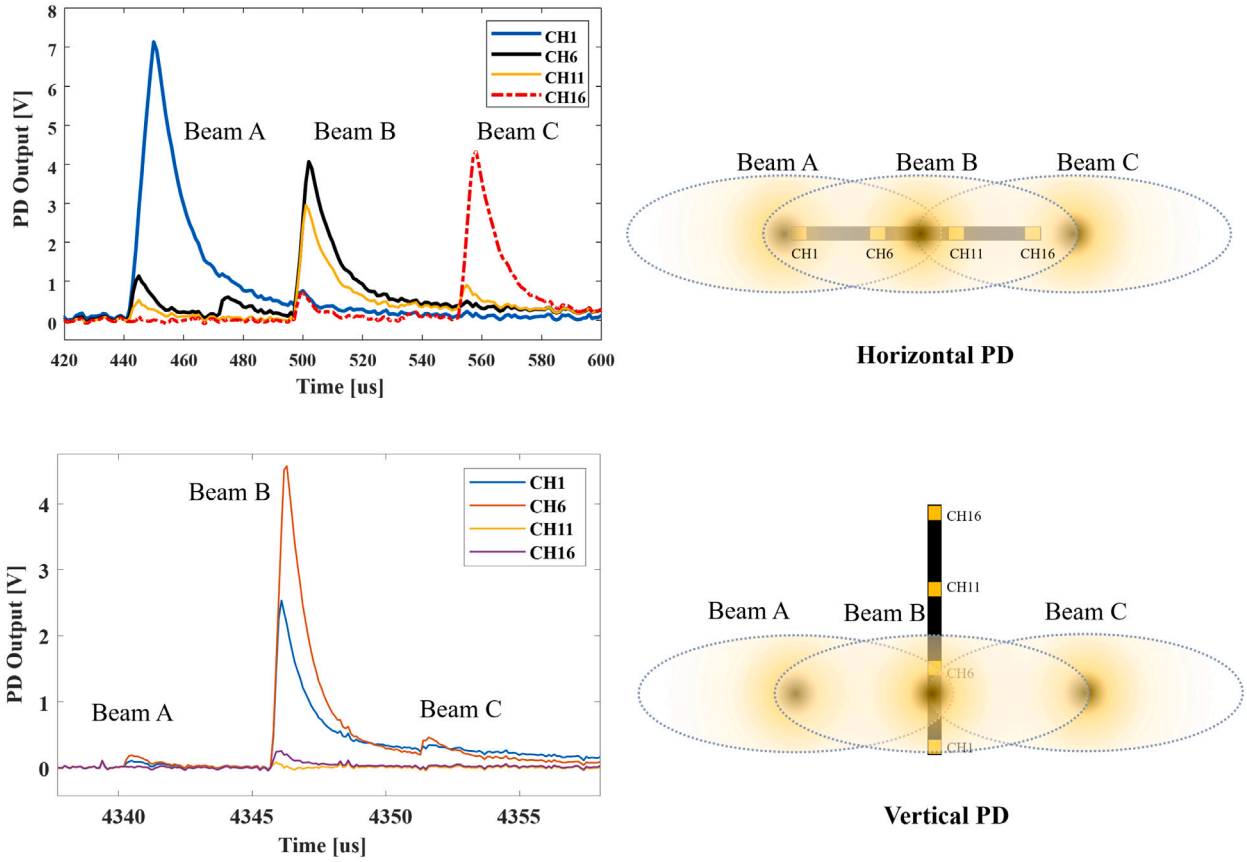


Fig. 6. Three beams (A to C) projected on the horizontally aligned (above) and vertical aligned (below) photodetector in consequent time that generate signals from each diode.

5.2. Pose of LiDAR sensor from vehicle

The LiDAR sensor alignment from the vehicle body (${}^V_L\mathbf{M} = [{}^V_L\mathbf{R} | {}^V_L\mathbf{T}]$) can be defined as

$${}^V_L\mathbf{M} = {}^V_O\mathbf{M} {}^O_L\mathbf{M} = {}^V_O\mathbf{M} [{}^O_L\mathbf{R} | {}^O_L\mathbf{T}] \quad (9)$$

where ${}^O_L\mathbf{M}$ is the Vehicle–Target pose and ${}^O_L\mathbf{M}$ is the Target–LiDAR sensor pose, as shown in Fig. 4.

Since this paper assumed that the position of the targetboard and the vehicle, ${}^O_L\mathbf{M}$, are predetermined in the factory environment, the calibration problem becomes estimating the Target–LiDAR sensor pose (${}^O_L\mathbf{M}$).

5.3. Relative pose of LiDAR sensor

The relative 6-DOF pose of the LiDAR sensor from the reference target frame can be estimated by solving for the transformation matrix ${}^O_L\mathbf{M}(\phi, \theta, \psi, \Delta x, \Delta y, \Delta z)$. To solve for the transformation matrix ${}^O_L\mathbf{M}$, the 3D positions of feature beam points measured by the target body frame (${}^O\mathbf{p}_i$) and measured in the LiDAR sensor frame (${}^L\mathbf{p}_i$) are necessary. With the proposed PD–target system, Eq. (9) is modified to include the photodetector coordinate frame, D.

$${}^D\mathbf{P} - \mathbf{P}_{offset} = {}^O_L\mathbf{M} {}^L\mathbf{P} = [{}^O_L\mathbf{R} | {}^O_L\mathbf{T}] {}^L\mathbf{P} \quad (10)$$

where

$${}^O_L\mathbf{R} = \mathbf{R}_z(\phi)\mathbf{R}_y(\theta)\mathbf{R}_x(\psi), \quad {}^O_L\mathbf{T} = [\Delta X, \Delta Y, \Delta Z]^T \quad (11)$$

The process of obtaining the position of the correspondence feature points, ${}^D\mathbf{P}$ and ${}^L\mathbf{P}$ are described in the following section. Note that since the signal processing for each PD alignment are similar, only the horizontal aligned photodetectors are explained in the section.

5.4. Estimating the position of the beam center

For each LiDAR sensor scan, there would be at most three beams projected on the same photodetector in the horizontal and vertical alignment, as shown in Fig. 6. With the given configuration of the target board at a distance of 2.5 m, the diameter of a beam spot would be 19.6 mm. The beams on the photosensitive area overlap with an interval of 9.7 mm but at different times with 55- μ s period.

The diode channel closest to the beam center outputs the highest peak level because the beam power intensity is approximately distributed as a bell-shaped. The first set of peaks is generated by *beamA*, and *CH1* showed a higher level than the others because it is the closest to the beam center. After 55 μ s from the first peaks, *beamB* is projected on the PD closest to *CH11*, which could be inferred by comparing the peak level among other channel outputs. Thus, by analyzing the distribution of the peak values for multiple beam projections, the position and the time of the beam scanning over the PD can be derived. Note that the beam intensity patterns should be analyzed for different types of LiDAR sensors.

From the three beams on the horizontal aligned PD, the feature beam spot is selected as the closest one to the PD center. The center position for each beam can be estimated by analyzing the output signals in the spatial domain. Starting at the origin of the PD coordinate frame on *CH1*, the center and the end (*CH16*) of the PD would be 7.5 mm and 15 mm, respectively. On this PD coordinate frame, the maximum output of each channel for a beam are plotted as shown in Fig. 7. The similar process is conducted for the vertical PD alignment to find the beam center.

The laser power spectrum was modeled as a Gaussian distribution [42], and the beam center is estimated by applying a Gaussian fit on the maximum signals of each PD channel. This paper used the iterative procedure of Gaussian fitting proposed in [43], which is

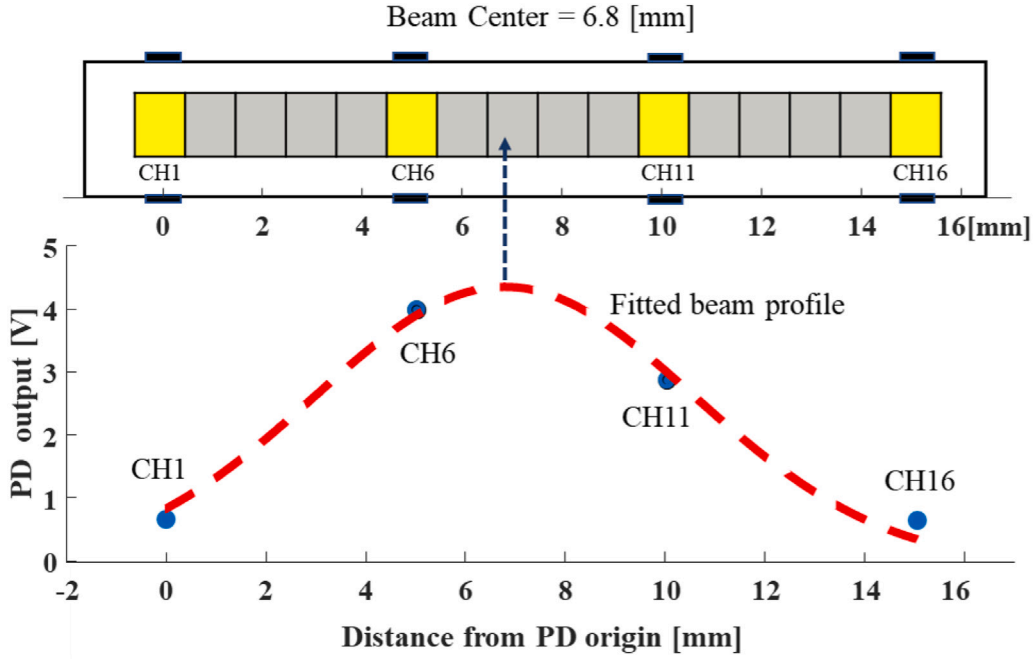


Fig. 7. Position of the beam center is estimated by applying a Gaussian fit on the signals of each diode channel.

known to be a robust and fast method. If the natural log is applied on both sides of 1D Gaussian model, then a function of the second-order polynomial can be derived.

$$\ln(y) = \ln(A) - \frac{(x-\mu)^2}{2\sigma^2} \quad (12)$$

$$= -\frac{x^2}{2\sigma^2} + \frac{2\mu x}{2\sigma^2} + \left(-\frac{\mu^2}{2\sigma^2} + \ln(A)\right) = a_2 x^2 + a_1 x + a_0$$

where x is the beam power input, y is the voltage output of the PD, μ is the average, or the beam center position, σ is the standard deviation and A is a weight value.

However, the actual output voltage signal, y , is corrupted with sensor noise. To apply the iterative procedure of Gaussian fitting [43] to achieve more robustness to the noise, Eq. (11) is modified as

$$\begin{bmatrix} \sum_{i=1}^m x_i^4 y_{i,(k-1)}^2 & \sum_{i=1}^m x_i^3 y_{i,(k-1)}^2 & \sum_{i=1}^m x_i^2 y_{i,(k-1)}^2 \\ \sum_{i=1}^m x_i^3 y_{i,(k-1)}^2 & \sum_{i=1}^m x_i^2 y_{i,(k-1)}^2 & \sum_{i=1}^m x_i y_{i,(k-1)}^2 \\ \sum_{i=1}^m x_i^2 y_{i,(k-1)}^2 & \sum_{i=1}^m x_i y_{i,(k-1)}^2 & \sum_{i=1}^m y_{i,(k-1)}^2 \end{bmatrix} * \quad (13)$$

$$\begin{Bmatrix} a_{2,(k)} \\ a_{1,(k)} \\ a_{0,(k)} \end{Bmatrix} = \begin{Bmatrix} \sum_{i=1}^m x_i^2 y_{i,(k-1)}^2 \ln(y_i) \\ \sum_{i=1}^m x_i y_{i,(k-1)}^2 \ln(y_i) \\ \sum_{i=1}^m y_{i,(k-1)}^2 \ln(y_i) \end{Bmatrix}$$

where k is the number of iterations, with a value of $k=10$ used in this study, and $y_{i,(k)}$ is defined as

$$y_{i,(k)} = \begin{cases} y_i & \text{for } k = 0 \\ \exp(a_{2,(k)} x_i^2 + a_{1,(k)} x_i + a_{0,(k)}) & \text{for } k > 0 \end{cases} \quad (14)$$

Finally, the model parameters μ and σ are then solved as

$$\sigma^2 = -\frac{1}{2a_{2,(k)}}, \quad \mu = a_{1,(k)}\sigma^2 = -\frac{a_{1,(k)}}{2a_{2,(k)}} \quad (15)$$

Since there were only four dataset measurements from the four laser scan channels, two more sets at -5mm and 20mm were augmented with the value of 0.1V , which is approximately the noise level.

After estimating the center position for all the beams on the PD, the feature beam point is then selected that has the closest value to the center or 7.5mm .

The next process is to match the corresponding beam point (${}^L P_i$) from the LiDAR sensor raw data that contains the 3D position of each

beam in polar or Cartesian coordinates with respect to the sensor body frame. To find the correspondence beam, the photodetector needs to be detected and segmented from the target surface by LiDAR sensor measurement. However, using only the 3D point data, the PD is not distinguishable from the target surface because the PD is too flat to show any clear depth differences.

Instead of using the LiDAR sensor 3D data, this paper used the difference in the reflectivity values of the beam on the PD from their neighbor beams. The surroundings of the PD are covered with a black colored background to force a lower reflectance intensity than that on the PD sensor surface [44].

Now, we have the matching pair of the position data of each key beam point, ${}^L P_i(\omega, \alpha, r)$ and ${}^O P_i(x, y, z)$, which have been obtained from the PD signals and the LiDAR sensor data, respectively.

The azimuth (α) from the point cloud and the matching beam center from the Gaussian fit is plotted for 50 LiDAR sensor scans to validate the correspondence matching. Given that the azimuth of the key beam point varies over the scans due to the LiDAR sensor rotation fluctuations, the beam center should be changed proportionally. However, the results showed some outliers deviated from the expected linear relationship, as plotted in Fig. 8. All outliers have the offset error of 0.2 degrees, which is the azimuth resolution of one beam index in the same vertical channel. The offset by one index could have been caused due to several factors, such as the potential for data loss of the first or the last beam of the target surface by the sharp edge of the target, which occasionally occurs.

At the experimental configuration of the target position, such an index offset can cause as high as an 9.7mm center position error.

The actual relationship between the azimuth and the center position should be a function of the tangent, but we approximated it as a linear curve within a small range of 1 azimuth degree. The outliers removed using RANSAC are shown as the solid red curve in Fig. 8. Thus, the pose estimation applied the new ${}^O P_x$ from this post-processing with all LiDAR sensor scan data as

$${}^D \mathbf{p}_{i,k} = \begin{bmatrix} \hat{\mu}_{i,k} = v_i + \tau_i \alpha_{i,k} \\ 0 \\ {}^D z_{i,k} \end{bmatrix} \quad (16)$$

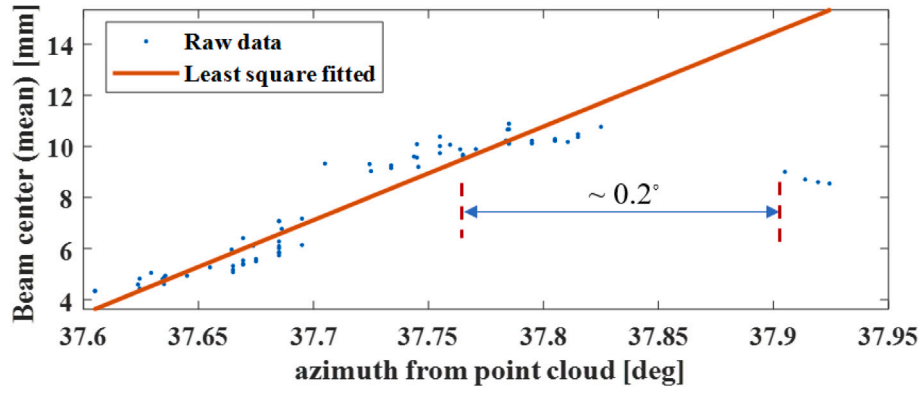


Fig. 8. The azimuth (α) from the point cloud data and the position of the beam center from the photodetector are plotted for 50 scans. The outliers are removed to obtain a linear relationship for correct correspondence matching.

5.5. Optimizing pose estimation

Now, the corresponding feature points have been obtained from the target board and the LiDAR sensor, and the relative pose of the LiDAR sensor with respect to the target body frame can be estimated. This paper applied the Levenberg Marquardt method for the iterative optimization to minimize the cost function of the pose error to obtain the best fit values for the pose variables β^* .

$$\beta^* = \arg \min_{\beta} \sum \left\| \mathbf{O} \mathbf{P} - [\mathbf{O} \mathbf{R}_i^{\mathbf{O}} \mathbf{T}_i^{\mathbf{O}}] \mathbf{L} \mathbf{P} \right\| \quad (17)$$

$$= \beta - \eta (\mathbf{J}^T \mathbf{J} + \lambda \text{diag}(\mathbf{J}))^{-1} \mathbf{J}^T \mathbf{F}(\beta)$$

where

$$\beta = (\phi, \theta, \psi, \Delta x, \Delta y, \Delta z) \quad (18)$$

Here, $\mathbf{J}$ is the Jacobian matrix for the cost function $\mathbf{F}$ in Eq. (19), a constant $\eta=0.02$, and the varying tuning rate λ is initially set at 0.3.

$$F(\beta)_i = \begin{bmatrix} \mathbf{O} P_{1,i} + r_i(c\alpha_i c\omega_i)(c\psi s\phi - c\phi s\psi s\theta) - r_i(s\alpha_i)(s\psi s\phi + c\psi c\phi s\theta) \\ -r_i(c\alpha_i s\omega_i)(c\theta c\phi) - \Delta x \\ r_i(s\alpha_i)(c\phi s\psi - c\psi s\theta s\phi) - r_i(c\alpha_i c\omega_i)(c\psi c\phi + s\psi s\theta s\phi) \\ -r_i(c\alpha_i s\omega_i)c\theta s\phi - \Delta y \\ \mathbf{O} P_{3,i} + r_i(c\alpha_i s\omega_i)s\theta - r_i(s\alpha_i)c\psi c\theta - r_i(c\alpha_i c\omega_i)c\theta s\psi - \Delta z \end{bmatrix} \quad (19)$$

6. Experiment

Experiments were conducted to evaluate the accuracy and precision of the proposed system with the designed test bench that consisted of the PD-target board and the LiDAR sensor pose controller module, as shown in Fig. 9. Since the previous methods require multiple target boards and are not applicable to this experiment setup, the experimental performance of the proposed system is compared indirectly in Table 1.

6.1. Test-bench setup

The target is a flat aluminum board with dimensions of 1 m in width and 0.54 m in height and has 1D PD arrays attached near each board corner. The board surface surrounding the photodiodes was covered with black papers to reduce the surface reflectance such that it was lower than the photodiode surface. The photodiodes were aligned vertically on the top-left and bottom-right corners and aligned horizontally on the other remaining corners. Two external DAQs of 4 ADC ports at a maximum of 4 MHz were used to capture the photodiode output voltage. Note that only two photodetectors were used at a time due to limited availability of the high-performance ADCs.

The LiDAR sensor (VLP-16) is placed on the 2-DOF pose controller module positioned at 2.5 m from the PD-target board. The pose controller module consisting of a rotation motor stage and a linear motor stage can control the precise yaw and x-position of the LiDAR sensor.

The yaw angle is controlled within 3 degrees of the range with a precision of 0.01 degree, and the horizontal position is controlled within 30 mm of the range with a precision of 0.01 mm.

6.2. Experimental results

The initial pose of the LiDAR sensor with respect to the target body frame is set as $X_O = -0.7$ m, $Y_O = -2.5$ m. The relative pose of the LiDAR sensor from the target board was estimated by the proposed algorithm at various reference poses of the yaw angle and x-position.

6.2.1. Yaw estimation test

In the first test, the yaw rotation of the LiDAR sensor was controlled from -3 to 3 degrees with a step of 0.5 degrees by the motor stage. For each reference yaw angle, 50 LiDAR sensor scans were repeated to evaluate the accuracy and precision of the relative pose estimation.

Fig. 10(a) shows the horizontally aligned PD-target system is tracking the input yaw angle with high accuracy. The yaw estimation error plotted in Fig. 10(b) indicates that the highest offset error is approximately 0.2 degrees at the reference angle of -3 degrees. The overall accuracy and precision of the yaw estimation are calculated to be 0.05 degrees.

The same test was repeated with the vertically aligned photodetector array and plotted in Fig. 10(c). The overall accuracy is under 0.03 degrees and the precision is under 0.06 degrees. The highest offset error is shown at the angle of 3 degrees, with the error value of approximately 0.1 degree.

6.2.2. Displacement estimation test

A similar test was conducted by varying the X-position of the LiDAR sensor from -30 mm to 30 mm with a step of 5 mm. Through the test, all orientation angles were fixed at the initial values. The estimation errors of the sensor position for both the horizontally and vertically aligned photodetectors are plotted in Fig. 11. As shown in the results, both alignment types can detect the LiDAR sensor offset displacement with an error of less than 3 mm.

Fig. 12 shows the statistics of the estimation error for all orientations and the displacement for this test. The horizontally and vertically aligned photodetectors have similar accuracy and precision, except the precision of the roll and tilt for the horizontal alignment seems slightly higher by 0.02 degrees.

The overall evaluation of the proposed system for horizontal and vertical aligned photodetector from averaging all test results as presented in Table 3. Considering the possibility of experimental error, both photodetector alignments are observed to yield a similar level of accuracy and precision for all orientations and displacements, which are within 0.1 degree and 3 mm. The alignment measurement results by the method based only the target board [9] for the same experiment

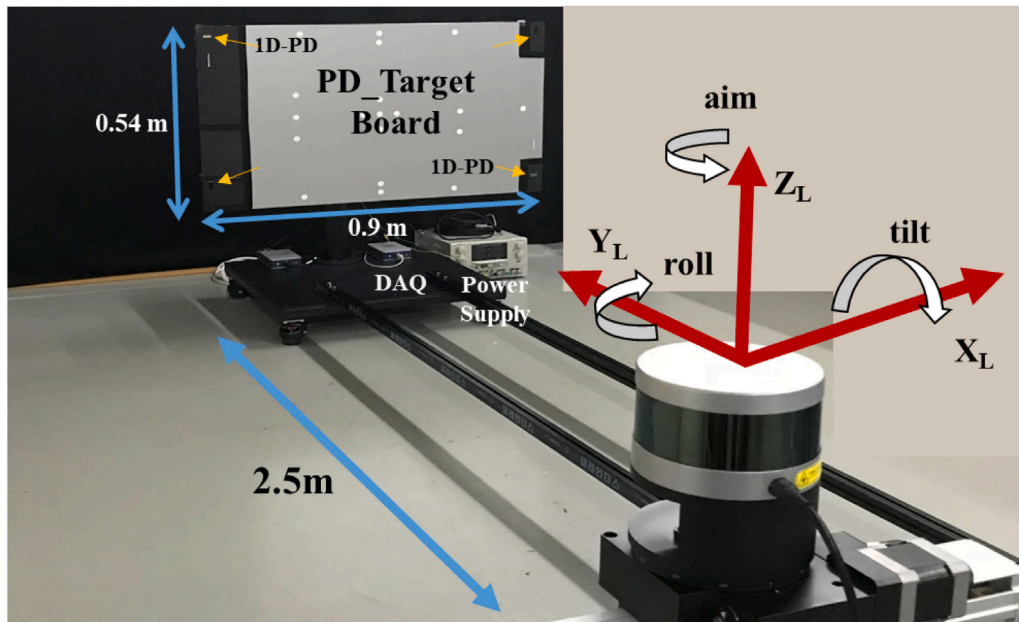


Fig. 9. Experimental setup consisting of the PD-target board and the 2-DOF motion stage to control the LiDAR sensor reference pose within a range of 3 degrees and 30 mm.

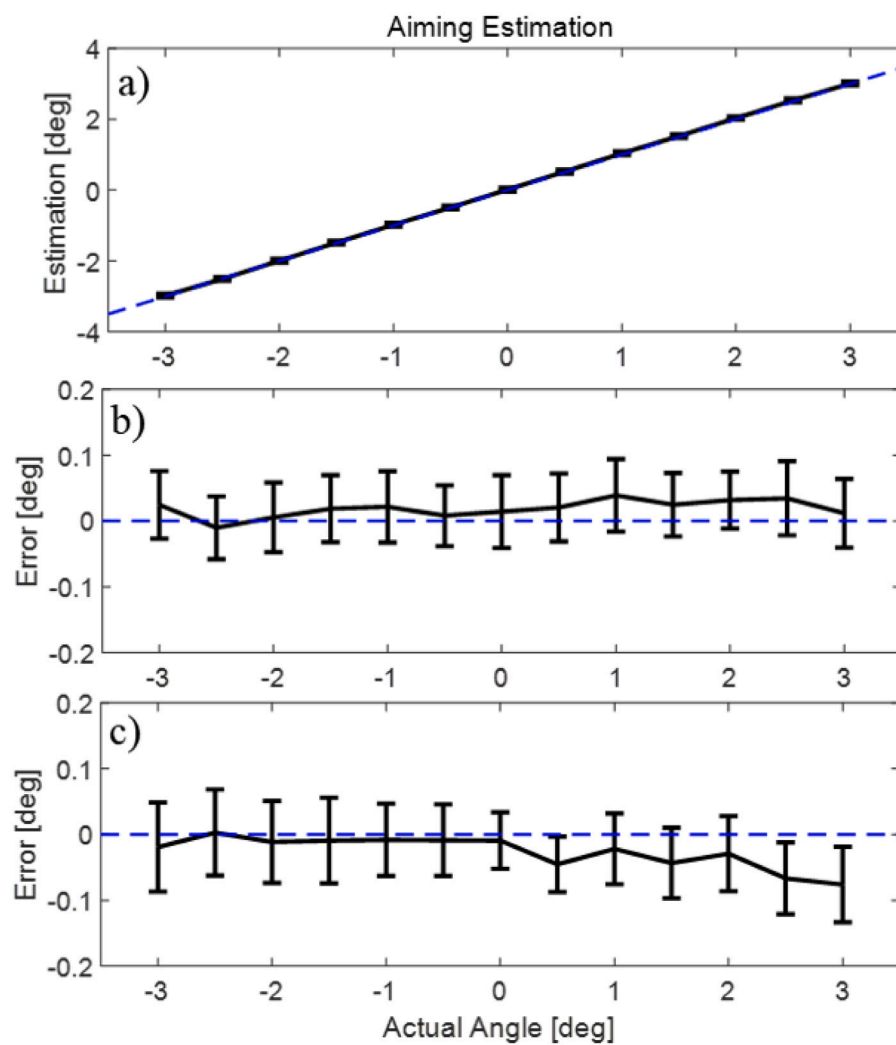


Fig. 10. The yaw estimation (a) tracking and error from -3 to 3 degrees (b) for horizontal aligned and (c) for vertical aligned PD.

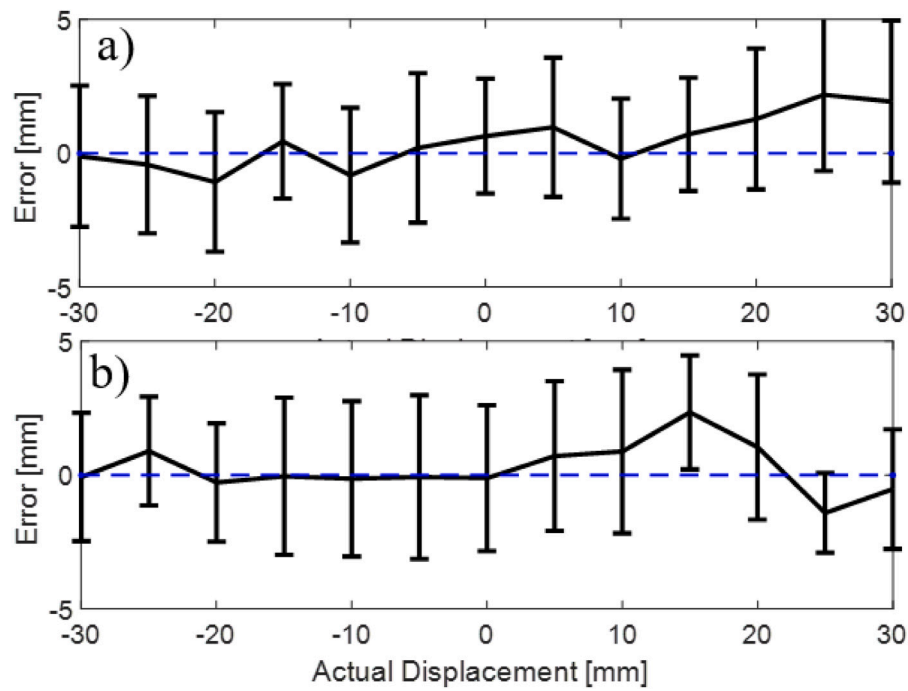


Fig. 11. The displacement estimation error statistics from 50 scans by (a) Horizontal aligned (b) vertically aligned photodetector.

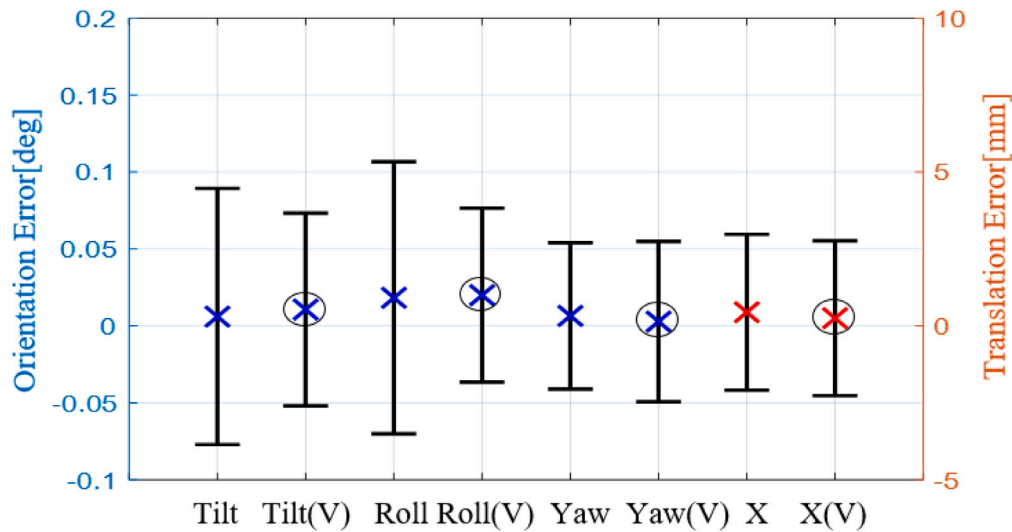


Fig. 12. Statistics of orientation angles and X-position errors for translation experiment. The circle mark indicates the results of the vertically aligned photodetectors.

were compared in Table 3. From the comparison, some improvement on the precision by the proposed method are shown clearly.

The given experiment setup of static method with a single target within a short range was not applicable for other previous methods. Also, they are hard to be compared directly. Nevertheless, previous methods are compared with the proposed system in Table 1 to provide a general overview of different approaches.

Compared to the high depth measurement noise and low resolution of a mobile LiDAR sensor, the 0.1 degree and 3 mm calibration errors are quite an improvement. If more photodetectors could be used, then the performance of the pose estimation would be improved more.

7. Conclusion

This paper proposed a novel PD–target calibration system for the extrinsic parameter calibration of LiDAR sensor mounted on a mobile

Table 3

Overall Experimental Result of Alignment Estimation.

Estimation		Tilt (°)	Roll (°)	Yaw (°)	ΔX (mm)
Horizontal PD	Accuracy	0.03	0.04	0.01	0.6
	Precision	0.11	0.10	0.05	2.3
	Accuracy	0.01	0.01	0.02	0.3
Vertical PD	Precision	0.08	0.08	0.05	2.6
Board only [9]	Accuracy	0.01	0.04	0.01	1.1
	Precision	0.19	0.09	0.11	4.9

platform for automatic sensor misalignment inspection that can be readily deployed in the actual in-factory vehicle production. The proposed system was designed to improve the one-shot calibration method based on a single board [9] by embedding photodetector sensors on the target surface to obtain the positions of laser beams more accurately.

With the assumption that relative position of the target and the vehicle frame are assumed to be known in the factory-level environment, this paper focused on estimating the relative pose of the LiDAR sensor from the target board.

The PD-target calibration system is composed of a planar target board with modules of NIR photodetector arrays attached near the corners of the board. With the photodetector arrays, the precise position of the laser beams on the board surface can be measured and used as the correspondence feature points for estimating the relative pose between the LiDAR sensor and the target frame body. The proposed system was evaluated in terms of the accuracy and precision of the pose estimation with the designed test bench that can control the reference yaw and horizontal position of the LiDAR sensor. Experimental results tested on VLP-16 LiDAR sensor showed that the system estimated the offset pose within 0.15 degrees and 3 mm of accuracy and precision combined.

There exist some technical issues that need to be overcome. This paper assumed that generally the LiDAR sensor is assembled properly in the factory-level and there is only a marginal misalignment that keeps the LiDAR sensor scan on the PD sensors. If the laser scan is off the PD, then the sensor assembly should be disqualified. However, for practical deployment, multiple arrays of photodetectors should be used to prevent the projection departure of laser beams. Also, the cost of high speed ADC DAQ boards with many channels is another limitation and this study only could test two photodetectors at a time. More implementations of photodetectors would improve the overall performance.

CRedit authorship contribution statement

Ji-Hwan You: Investigation, Methodology, Software. **Seontaek Oh:** Methodology, Software. **Jaeeun Park:** Investigation, Software. **Hyeongseok Song:** Investigation. **Young-Keun Kim:** Supervision, Conceptualization, Methodology, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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